

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE
AND EXPERT SYSTEMS

COURSE CODE: MLA0103

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)*

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I M. Raghavendra/192325022 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	

Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	Manufacturing	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of AI Application	Provide the name of the latest AI application selected and the corresponding AI domain.
2	Domain Selection	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	Introduction and Background	Describe the AI application and provide an overview of its purpose and importance.
4	Problem Identification	Explain the real-world problem or challenge addressed by this AI application.
5	Need for AI Solution	Analyze why AI is required and how it improves traditional approaches.
6	AI Technologies Used	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	Working Mechanism	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	Data Sources and Processing	Describe the type of data used and how the data is processed for intelligent decision-making.
9	Real-World Implementation	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	Impact and Benefits	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	Challenges and Limitations	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	Future Scope and Emerging Trends	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	Conclusion	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	References	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

Scenario-Based Research Assessment — Answer Script

Course Outcome: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems

Question Attempted: Q1 — Healthcare Domain

1. Title of AI Application

AI-Based Radiology Diagnostic System — Deep Learning for Medical Image Analysis (Domain: Healthcare — AI-Assisted Medical Diagnosis).

2. Domain Selection

Domain: Healthcare. The application researched is an AI-powered radiology diagnostic system that analyzes medical images (X-rays, CT scans, and MRIs) to detect abnormalities such as fractures, tumors, pneumonia, and other pathologies, assisting radiologists in faster and more accurate diagnosis.

3. Introduction and Background

Medical imaging is central to modern diagnosis, but the volume of scans generated daily far outpaces the number of available radiologists, especially in under-resourced hospitals. AI-based radiology systems use deep learning, particularly Convolutional Neural Networks (CNNs), trained on large annotated image datasets to automatically detect, localize, and classify abnormalities. Systems such as Google DeepMind's mammography AI, Aidoc's CT triage platform, and Qure.ai's chest X-ray analysis tool are already deployed in hospitals worldwide. Their purpose is not to replace radiologists but to act as a "second reader" — flagging urgent cases, reducing missed findings, and cutting reporting time.

4. Problem Identification

- Radiologist shortage: many hospitals, especially in rural or developing regions, have very few qualified radiologists relative to patient load, causing diagnostic backlogs.
- Human fatigue and error: manually reviewing hundreds of scans a day increases the risk of missed or delayed detection of critical findings (e.g., a small lung nodule or an early-stage fracture).
- Time-critical cases: conditions such as brain hemorrhage or pulmonary embolism require rapid identification; delays in reading scans directly affect patient outcomes.
- Inconsistent diagnosis quality: interpretation can vary between radiologists (inter-observer variability), affecting diagnostic reliability.

5. Need for AI Solution

Traditional (non-AI) workflows rely entirely on manual visual inspection, which does not scale with the growing volume of imaging data and cannot guarantee consistent, round-the-clock coverage. AI improves on this by:

- Processing scans in seconds, enabling automatic triage so the most urgent cases are reviewed first.
- Providing a consistent second opinion that does not suffer from fatigue, helping catch findings a tired or overloaded radiologist might miss.
- Learning from millions of prior labeled scans — a scale and consistency of pattern recognition no individual clinician can match.
- Freeing radiologists' time from routine, high-volume normal-scan screening to focus on complex, ambiguous cases.

6. AI Technologies Used

AI Technique	Role in the System
Convolutional Neural Networks (CNN)	Core deep learning architecture for extracting spatial features (edges, textures, shapes) from medical images and classifying abnormalities.
Computer Vision	Image pre-processing, segmentation of organs/regions of interest, and object localization (e.g., bounding boxes around a nodule).
Transfer Learning	Pre-trained models (ResNet, DenseNet, EfficientNet) fine-tuned on smaller labeled medical datasets to overcome limited data availability.
Explainable AI (Grad-CAM / saliency maps)	Generates heat-maps showing which regions of the image influenced the model's decision, building radiologist trust.
Natural Language Processing (NLP)	Used in some systems to auto-generate structured radiology reports from the model's findings.

7. Working Mechanism

The system follows a structured image-to-diagnosis pipeline. A medical scan is first pre-processed (noise removal, normalization, resizing) so that the input is standardized. A CNN then extracts hierarchical features — early layers detect edges and textures, deeper layers detect complex structures such as organ shapes or lesions. A classification/segmentation head produces a probability score for each possible condition and, where relevant, marks the exact region of concern. Post-processing generates a confidence score and an explainability heat-map, and the final output — a flagged finding with supporting visualization — is presented to the radiologist, who makes the final clinical decision. Feedback from radiologist corrections is often looped back to retrain and improve the model over time.

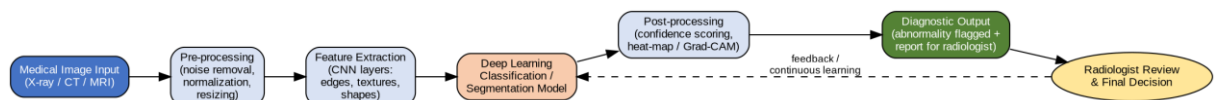


Figure 1: Working mechanism of an AI-based radiology diagnostic pipeline

8. Data Sources and Processing

Training data consists of large, de-identified medical imaging datasets (e.g., chest X-ray datasets such as ChestX-ray14, CT/MRI datasets from hospital partnerships, or public challenge datasets), each labeled by expert radiologists. Data processing involves anonymization to protect patient privacy (removing identifying metadata), image normalization and augmentation (rotation, contrast adjustment) to improve model robustness, and careful class-balancing since abnormal cases are typically far rarer than normal scans. Data is split into training, validation, and test sets, with test performance validated against radiologist-confirmed ground truth before clinical deployment.

9. Real-World Implementation

Organization / Product	Application
Google DeepMind / Google Health	AI system for breast cancer screening from mammograms, shown in studies to reduce false negatives and false positives compared to a single radiologist.
Aidoc	FDA-cleared AI platform integrated into hospital PACS systems to triage CT scans for conditions like intracranial hemorrhage and pulmonary embolism in real time.
Qure.ai	Deep-learning chest X-ray analysis widely deployed in India and other countries for TB and lung-abnormality screening, including in resource-limited settings.
PathAI / Paige (Pathology)	Related AI-assisted diagnosis in digital pathology, extending similar CNN-based image analysis to tissue slides for cancer detection.

10. Impact and Benefits

- **Efficiency:** reduces average scan turnaround time, enabling faster triage of critical cases.
- **Accuracy:** studies report AI-assisted reading can reduce missed findings and, in some screening tasks, match or exceed single-reader radiologist accuracy when used as a second reader.
- **Cost:** automating first-pass screening of routine, normal scans reduces radiologist workload and associated staffing costs, particularly valuable in low-resource settings.
- **Accessibility:** brings specialist-level screening capability to regions and clinics without on-site radiologists.

11. Challenges and Limitations

- **Data bias:** models trained mostly on data from certain populations or scanner types may perform less reliably on underrepresented groups or equipment, raising fairness concerns.
- **Regulatory and liability issues:** unclear accountability when an AI-assisted diagnosis is wrong — who is responsible, the developer or the clinician.

- Privacy and security: medical imaging data is highly sensitive; breaches or misuse of patient data carry serious ethical and legal consequences.
- Explainability: deep learning models are often "black boxes," making it hard for clinicians to fully trust or audit a specific decision without tools like Grad-CAM.
- Integration challenges: deploying AI into existing hospital IT/PACS workflows requires significant technical and change-management effort.

12. Future Scope and Emerging Trends

- Multimodal AI combining imaging with patient history, lab results, and genomics for more holistic diagnosis.
- Federated learning, allowing hospitals to collaboratively train models on decentralized data without sharing raw patient images, improving privacy and generalizability.
- Generative AI for synthetic medical image generation to augment rare-condition datasets and improve model robustness.
- Real-time AI integrated directly into imaging hardware for instant point-of-care screening in ambulances and rural clinics.
- Stronger regulatory frameworks (e.g., evolving FDA/CE guidelines) for continuous post-deployment monitoring of clinical AI systems.

13. Conclusion

AI-based radiology diagnostic systems demonstrate how deep learning and computer vision can directly address a pressing real-world problem — the shortage and workload of radiologists — by acting as an accurate, consistent second reader for medical images. While challenges around bias, explainability, and regulation remain, real-world deployments such as Aidoc and Qure.ai show measurable gains in efficiency and diagnostic support. As multimodal and federated learning techniques mature, such systems are likely to become an increasingly integral, trusted part of clinical workflows rather than a replacement for human expertise.

14. References

- Google Health, "AI system for breast cancer screening," *Nature*, 2020 — study on mammography AI performance.
- Aidoc, "AI-based triage and notification platform for radiology," company clinical documentation and FDA clearance summaries.
- Qure.ai, "Deep learning for chest X-ray interpretation," published validation studies and deployment case reports.
- Esteva, A. et al., "A guide to deep learning in healthcare," *Nature Medicine*, 2019.
- Topol, E., "Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again," 2019.